



PecoDrop Kiosk Operation Manual

پیکوڈراپ ری سائیکلنگ کیوسک — شہری و صارف گائیڈ

Comprehensive step-by-step operating guidelines for citizens and recyclers. Learn how to initiate hybrid touchless sessions, deposit plastic bottles, aluminum cans, and tetra paks, track multi-sensor acceptance, earn instant reward points, and sync with your mobile wallet.

✦ HYBRID TOUCHLESS START

Scan dynamic on-screen QR Code via PecoDrop App or press Key '0' on the tactile keypad.

🗑️ MULTI-MATERIAL CHUTE

Intelligent sorting for Plastic Bottles (PET), Aluminum Cans, and Tetra Pak beverage cartons.

🏆 1.15X ECO LEADERBOARD

Expanded dual-screen landscape display showcasing top recyclers, avatars, live points, and eco-impact.

MACHINE IDENTIFIER

PECO-2026 (Model K26)

INSTALLED LOCATION

Katra Neem Wala, Lahore

GPS COORDINATES

31.5826° N, 74.3276° E

DOCUMENT ID

DOC-PECO-UM-2026-V3.5

Manual Directory & Hybrid Quick Start

A concise operational index and visual 60-second walkthrough for all users and citizens.

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HYBRID QUICK START (60 SECONDS)

- 1 Start Session:** Scan on-screen QR with PecoDrop App OR press tactile Key '0' on machine.
- 2 Insert Items:** Insert empty Plastic Bottle, Aluminum Can, or Tetra Pak bottom-first into the illuminated chute.
- 3 Smart Classification:** Machine reads barcode, tests material via inductive sensors & ultrasonic profiling.
- 4 Flap Actuation:** Motorized servo flap opens 180° and deposits the container safely into the internal bin.
- 5 Touchless Claim:** Scan the dynamic wallet QR code on screen to claim points instantly, or type phone number on keypad.
- 6 Rating:** Rate your experience (1–5 Stars) to conclude the session.

اردو فوری رہنمائی (فوری استعمال کا طریقہ)

- ۱۔ سیشن کا آغاز: موبائل ایپ سے اسکرین پر موجود کیو آر کوڈا سکین کریں یا کی پیڈ پر '0' کا بٹن دبائیں۔
- ۲۔ بوتل یا کین ڈالیں: خالی پلاسٹک کی بوتل، کین، یا ٹیٹرا پیک نچلے حصے کی طرف سے اندر داخل کریں۔
- ۳۔ خودکار تصدیق: مشین سینسرز اور کیمرے کی مدد سے بار کوڈ اور میٹریل کی تصدیق کرے گی۔
- ۴۔ پوائنٹس وصولی: سیشن مکمل ہونے پر اپنے موبائل سے کیو آر کوڈا سکین کر کے فوری ریوارڈ پوائنٹس حاصل کریں۔

1. System Overview & Depository Architecture

Understanding the PecoDrop Kiosk external interface, smart deposit chute, and sensing capabilities.

Depository Anatomy & Hardware Components

The **PecoDrop Kiosk** is engineered for high-throughput citizen recycling. It features a rugged industrial chassis, an ultra-responsive multi-sensor chamber, and a motorized intake hatch that provides active anti-cheat protection.

COMPONENT	FUNCTION & SPECIFICATION
Intake Chute	Circular illuminated opening (Ø 125mm) with internal optical safety barrier.
Material Sensors	Inductive metal sensor (cans) + Ultrasonic height/depth profiler (bottles/tetras).
Servo Drop Flap	Motorized 180° swing flap that validates weight and drops accepted containers.
Dual Display Modes	Supports Portrait (62/38 split) and Landscape (0.85/1.15 split) display orientations.
Keypad & QR Engine	12-key tactile matrix alongside dynamic SVG QR engine for 100% touchless flow.

Accepted Multi-Material Containers

1. Plastic PET Bottles (پلاسٹک کی بوتلیں)

Water, soft drink, and juice bottles from 250ml to 2.5 Liters. Must be empty with intact readable barcode label.

2. Aluminum Beverage Cans (مشروبات کے کین)

Standard 250ml, 330ml, and 500ml beverage cans. Detected via inductive metal sensors. Must not be heavily crushed.

3. Tetra Pak Cartons (ٹیٹرا پیک ڈبے)

Flavored milk and fruit juice cartons. Classified via geometric volume and ultrasonic depth measurements.

Prohibited Items: Glass bottles, liquid-filled containers, aerosol spray cans, and general trash will be instantly rejected by the optical classifier.

Smart Safety Note: The machine entrance servo flap automatically locks if hands or foreign objects remain inside the chute. Never insert hands past the protective rim.

حفاظتی نوٹ: بوتل ڈالتے وقت ہاتھ مشین کے اندر مت داخل کریں۔ کسی بھی ہنگامی صورت میں مشین کا خود کار حفاظتی نظام کام روک دیتا ہے۔

2. Welcome Screen & Citizen Dashboard

Real-time diagnostics, dynamic touchless QR start card, and digital signage presentation.

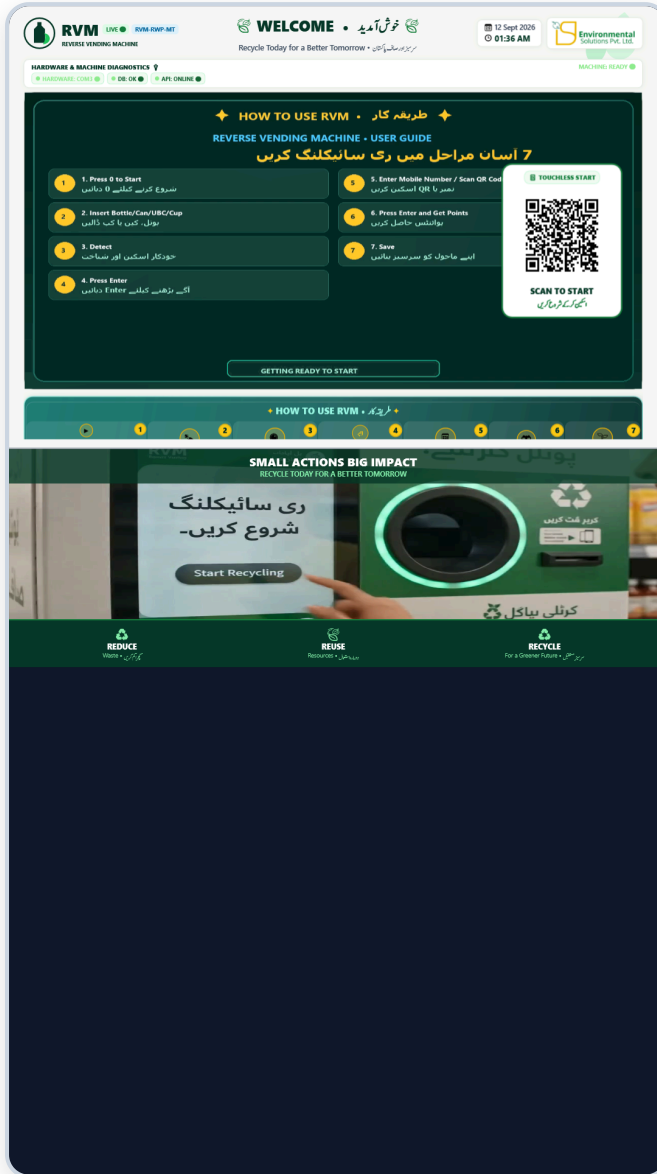


FIGURE 4.1 • PORTRAIT DISPLAY INTERFACE

The main standby screen greets citizens with real-time hardware health, an animated deposit guide, dynamic start QR code, and public eco-signage.

اسکرین پر خوش آمدید کا پیغام، مشین کی حالت اور سیشن شروع کرنے کے لیے کیو آر کوڈ اور بٹن کا آپشن موجود ہے۔

INTERACTIVE SCREEN ZONES (62% / 38% DUAL SPLIT)

- Top Diagnostics Bar:** Shows live machine status (MACHINE: READY ●, HARDWARE: COM7 ●, DB: OK ●, API: ONLINE ●).
- Touchless QR Start Card:** Large dynamic QR code (SCAN TO START / اسکن کریں) for instant contactless session startup via the PecoDrop Mobile App.
- Tactile Start Button:** Key 0 on the physical keypad allows tactile start without a smartphone.
- Live Recycling Counter:** Displays today's total recycled units and cumulative eco-points distributed to citizens.
- Bottom 38% Signage Area:** Seamlessly plays environmental awareness videos, public service campaigns, and sponsor broadcasts.

DIAGNOSTIC INDICATOR

OPERATIONAL MEANING

MACHINE: READY ●	Chute door unlocked, intake motors armed, system ready for deposit.
HARDWARE: COM7 ●	Microcontroller sensors, IR beams, and servos communicating at 9600 baud.
DB: OK ●	Local database operational, offline failover buffer active.
API: ONLINE ●	Connected to Central Cloud at isprvm.binishaqsoft.com for instant points sync.

2. 1-Minute Inactivity Idle Screen

Automatic power management, 50% promotional expansion, and instant touchless wake-up.

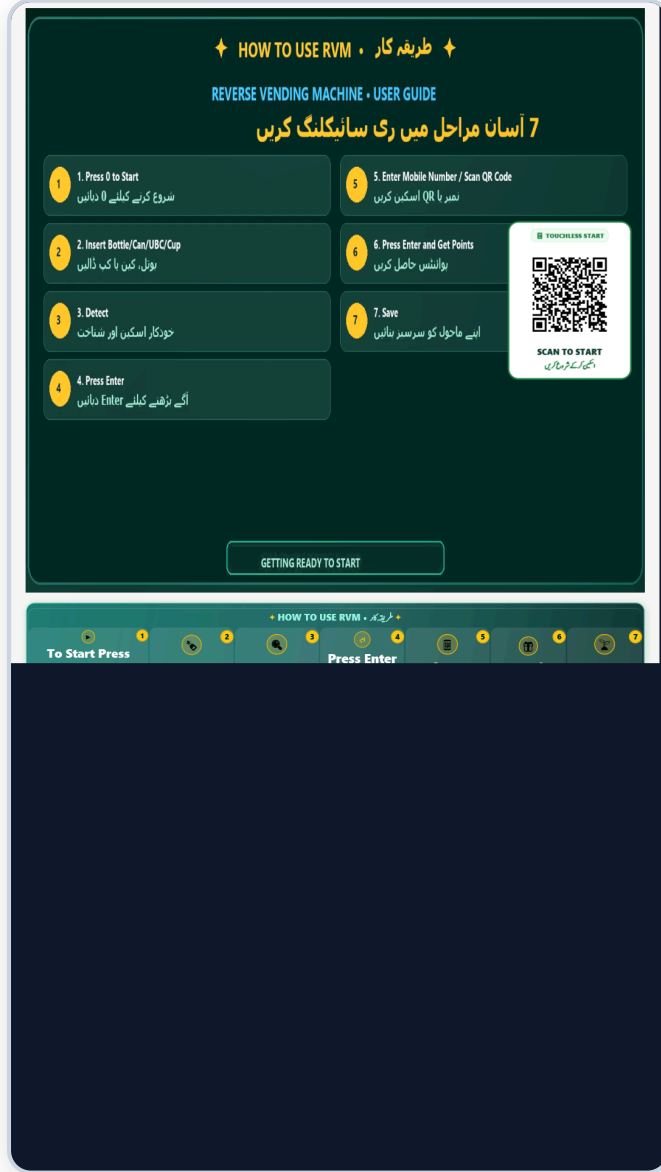


FIGURE 5.1 • EXPANDED IDLE STATE

When inactive for 60 seconds, the kiosk automatically transitions into an energy-efficient expanded instructional display.

ایک منٹ تک استعمال نہ ہونے پر اسکرین خود کار طریقے سے تعلیمی ویڈیو اور اشتہارات کو بڑا کر دیتی ہے۔

⚙️ IDLE MODE ARCHITECTURE & BEHAVIORS

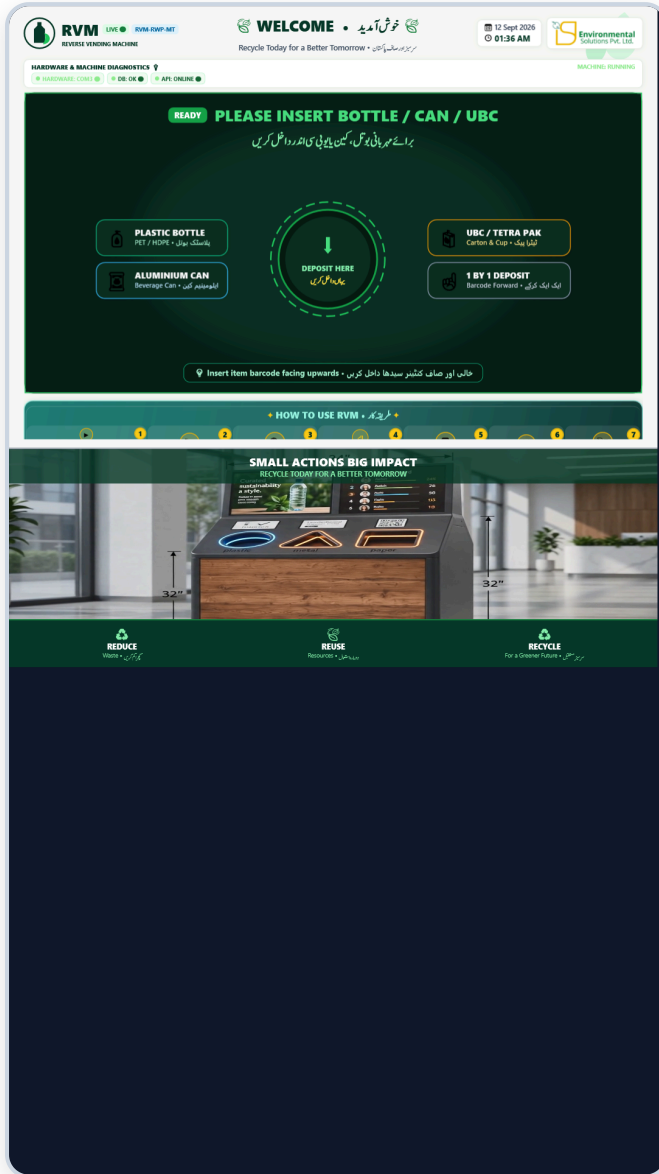
- **50% Height Expansion:** The promotional and instructional video window expands to 50% screen height to capture passerby attention.
- **Instant Touchless Wake-Up:** Scanning the floating on-screen QR code with your mobile app instantly wakes the kiosk and unlocks the session.
- **Hardware Wake-Up:** Pressing tactile Key 0 or approaching within sensor range immediately restores the active deposit UI.
- **Eco-Power Saver:** Chute illumination dims to standby levels to conserve electrical energy while maintaining sensor readiness.

💡 **Citizen Convenience Tip:** You do not need to wait for the video to finish. Simply scan the QR code with your phone camera or PecoDrop App, and the machine will instantly return to the active deposit screen.

فوری بیداری کا طریقہ: جیسے ہی آپ اپنا فون اسکرین کے کیو آر کوڈ کے سامنے لائیں گے، مشین فوری طور پر بولتلیں جمع کرنے کے لیے تیار ہو جائے گی۔

Step 1: Session Initiation (Touchless QR or Keypad)

Choose between 100% contactless mobile app interaction or tactile hardware keypad start.



STEP 1 OF 7 • INITIATION MODES

MODE A: TOUCHLESS QR SCAN

1. Open PecoDrop Mobile App on your phone.
2. Point your camera at the on-screen SCAN TO START QR code.
3. Kiosk beeps, links your profile, and unlocks the deposit door instantly.

MODE B: TACTILE KEYPAD '0'

1. Stand in front of the kiosk.
2. Press Key 0 on the rugged metallic keypad.
3. Chute lights green and session timer begins for guest recycling.

Session Lock & Security: Once a session is initiated, the kiosk is exclusively bound to the depositor. Other inputs are temporarily locked until the session completes.

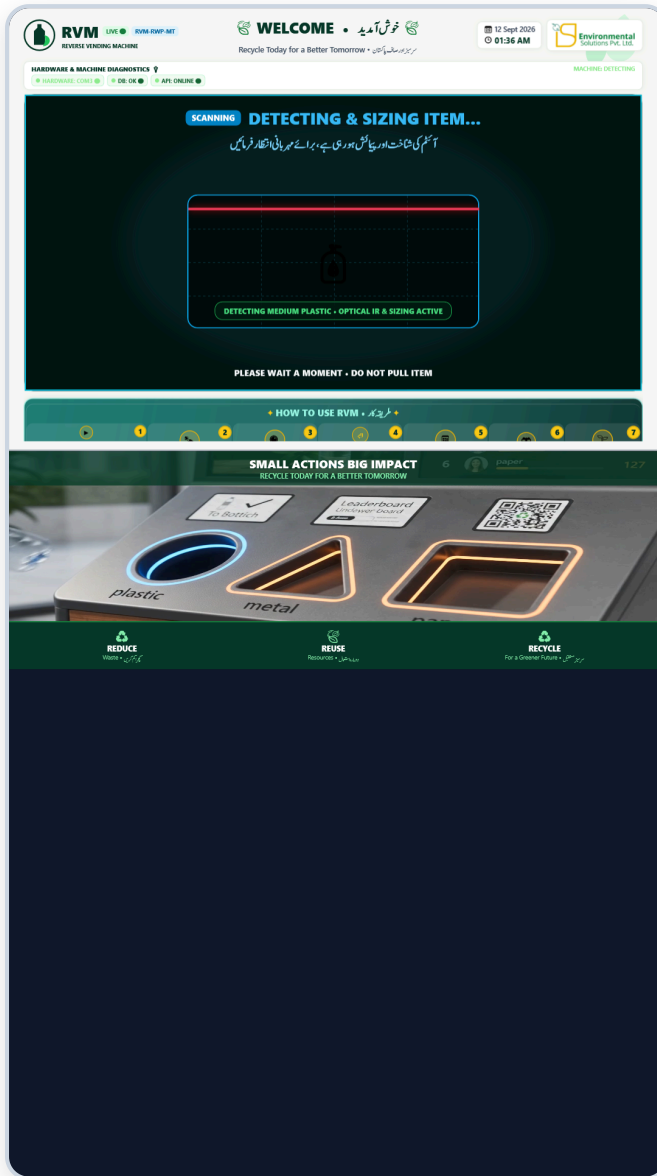
سیشن کا آغاز: جب سیشن شروع ہوتا ہے تو مشین کا دروازہ کھل جاتا ہے اور اسکرین پر بوس ڈالنے کی ہدایات ظاہر ہوتی ہیں۔

SESSION RULES & SAFETY CHECKS

- **Session Timeout:** Each deposit step gives 60 seconds before auto-closing. Inserting an item resets the timer.
- **Door Indicator:** The LED halo around the intake chute transitions from Pulsing Amber to Solid Emerald Green.
- **Cancellation:** Press Key * (Cancel) anytime before inserting bottles to end the session safely.

Step 2: Container Insertion & Multi-Material Sizing

Proper orientation, material classification rules, and sensor sizing detection.



STEP 2 OF 7 • INSERTION GUIDE

Insert your container **bottom first** into the illuminated chute with the barcode facing upward. Keep hands outside the safety zone.

بوتل یا کین کو نچلے حصے کی طرف سے اندر ڈالیں تاکہ بار کوڈ کیمرے کے سامنے آئے۔

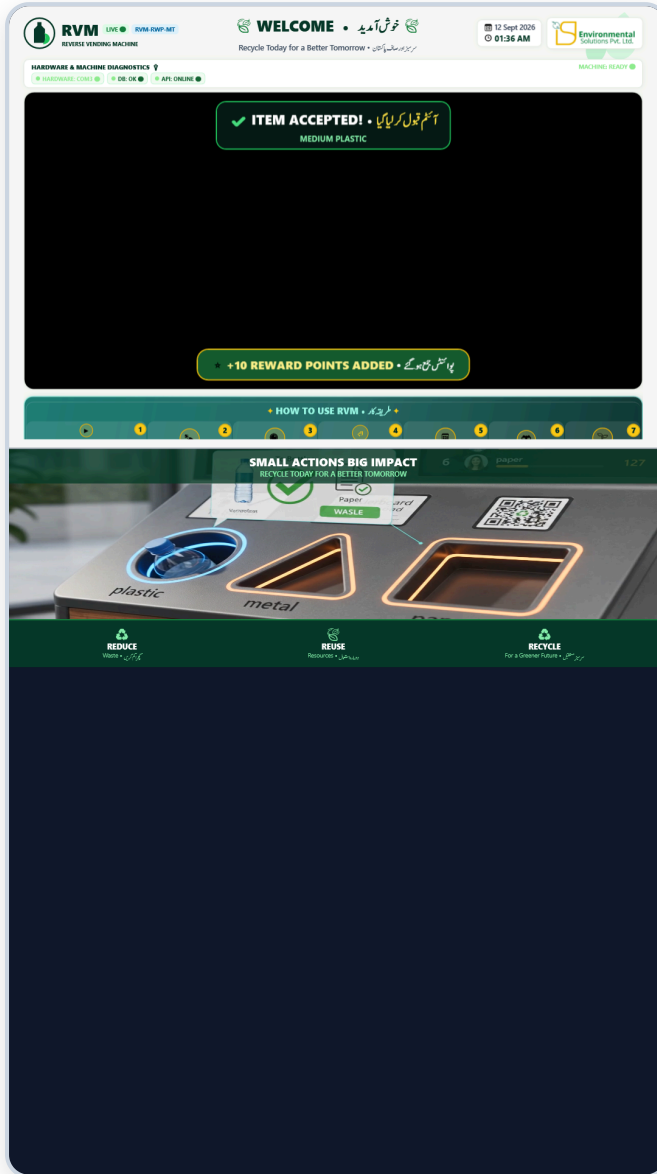
MULTI-MATERIAL ACCEPTANCE GUIDELINES

CONTAINER TYPE	ACCEPTANCE CRITERIA	REWARD
PET Bottle (Small)	250ml – 500ml, empty, cap on/off, label readable.	+5 Points
PET Bottle (Large)	1.0L – 2.5L, clean, barcode intact.	+10 Points
Aluminum Can	250ml – 500ml beverage can, not flattened.	+15 Points
Tetra Pak Carton	200ml – 1000ml juice/milk carton, straw removed.	+8 Points

⚠ Avoid Rejection: Ensure containers are completely drained of liquids. Residual liquids trigger weight imbalance sensors and will cause immediate ejection.

Step 3: Interactive Sensing States & Overlays

Real-time classification, audio-visual feedback, and animated celebration characters.



STEP 3 OF 7 • SENSING & FEEDBACK

The kiosk triggers high-speed optical scanning and displays delightful animated characters celebrating your eco-contribution.

بوتل کی کامیاب جانچ پر اسکرین پر خوشگوار انیمیشن اور پوائنٹس کا اضافہ دکھائی دیتا ہے۔

EXCLUSIVE PECODROP ANIMATED FEEDBACK

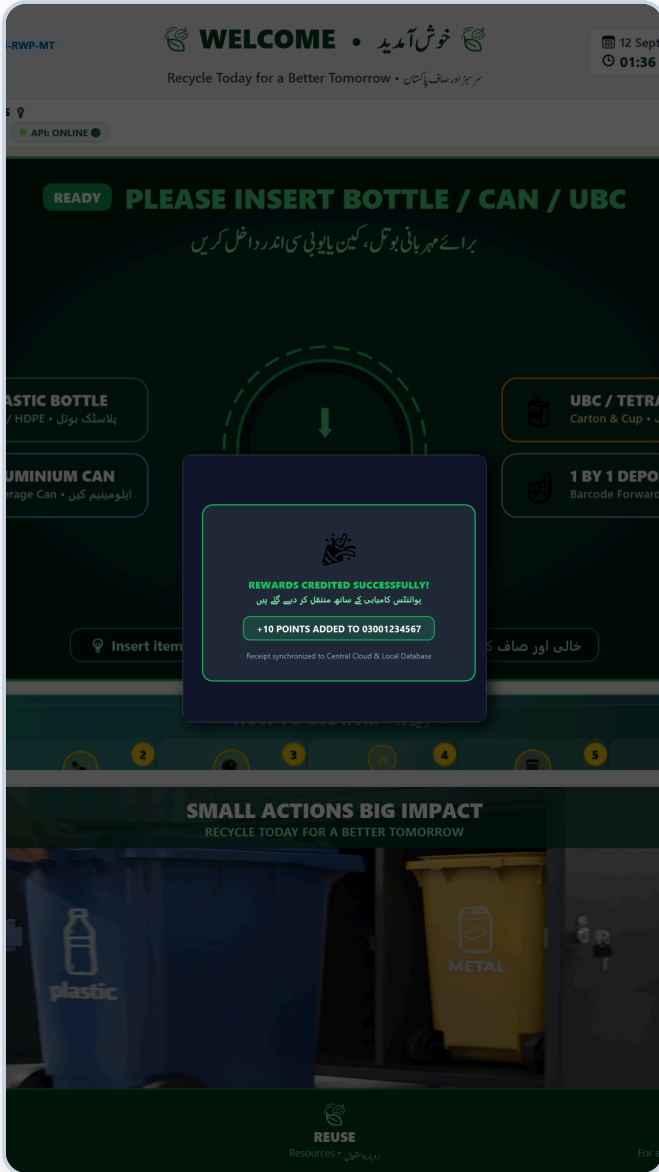
- **DancingPlastic** (پلاسٹک رقص): An energetic green plastic character pops up celebrating successful bottle deposit.
- **DancingCan** (کین اینیمیشن): A vibrant metallic character dancing to celebrate metal recycling.
- **DancingTetra** (ٹیٹرا اینیمیشن): A friendly juice box character congratulating the citizen.
- **Audio Confirmation**: A clean, pleasant chime confirms successful acceptance, followed by an Urdu voice prompt: "شکریہ! آپ کے ہوانٹس شامل کر دیے گئے ہیں".

✗ Rejection Overlay (مسترد ہونے کی صورت میں)

If an unaccepted item (glass, paper cup, crushed can) is inserted, the chute lights pulse red, the screen displays a warning overlay, and the item is safely returned.

Step 4: Drop Confirmation & Anti-Cheat Validation

Motorized 180° servo gate actuation, optical trap door, and fraud prevention.



STEP 4 OF 7 • DROP GATE ACTION

Once verified, the motorized servo gate swings downward 180°, dropping the container into the internal collection bin while secondary optic beams confirm transit.

تصدیق کے بعد اندرونی گیٹ کھلتا ہے اور بوتل خود کار طریقے سے محفوظ اسٹوریج میں چلی جاتی ہے۔

ANTI-CHEAT & TAMPER PREVENTION SAFEGUARDS

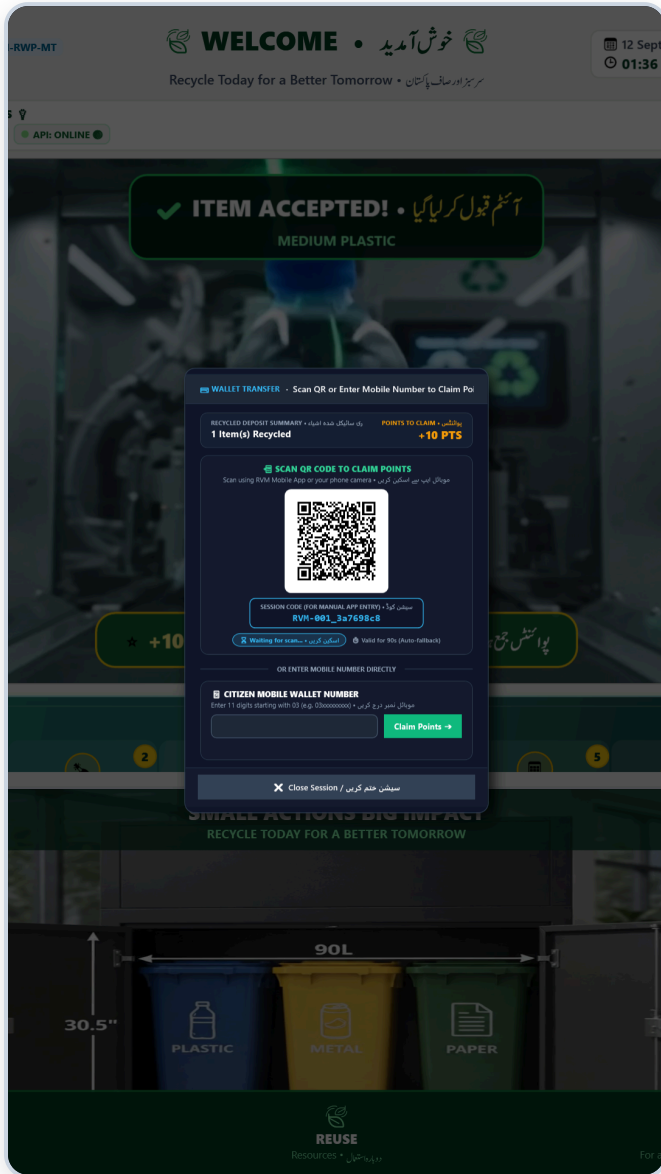
- **Anti-Fish Barrier:** A dual-tooth mechanical barrier prevents retrieving bottles with attached strings or wires.
- **Secondary Optic Gate:** Bottom optical receivers verify that the bottle fully passed into the bin before crediting points.
- **Servo Torque Limiting:** If resistance is detected during gate movement, the servo reverses instantly for safety.
- **Continuous Deposit:** Citizens can immediately insert their next bottle without pressing any buttons.

Session Tally Counter: The top right of the screen updates in real time, showing:

• Total Bottles: 4 | Total Cans: 2 | Total Tetra: 1 | Earned Points: 58

Steps 5 & 6: Touchless QR Wallet Scan & Reward Sync

Real-time reward tally, instant mobile QR claim, or 11-digit keypad entry fallback.



STEPS 5 & 6 OF 7 • CLAIM REWARDS

Press **FINISH** / ختم کریں (Key '#') to generate your dynamic reward claim QR code on screen.

پوائنٹس وصول کرنے کے لیے اسکرین پر ظاہر ہونے والے کیوار کوڈ کو اپنے فون سے اسکین کریں۔

⚡ TOUCHLESS CLAIM (RECOMMENDED)

1. Open PecoDrop App Wallet screen.
2. Scan the dynamic on-screen session QR.
3. Points transfer instantly to your citizen balance with instant push notification.

☎ PHONE KEYPAD FALLBACK

1. Enter your 11-digit mobile number (e.g. 03001234567) on the physical keypad.
2. Press # to confirm.
3. SMS receipt is sent with your point balance.

🎁 REWARD REDEMPTION ECOSYSTEM

- **Utility Bill Discounts:** Redeem points for electricity, gas, or municipal water credits.
- **Public Transport Vouchers:** Convert points to Metrobus and Orange Line transit passes.
- **Eco-Store Coupons:** Get food and grocery discount coupons at partner retail outlets.

Step 7: Citizen Rating & Session Complete

5-star citizen satisfaction rating, municipal service telemetry, and auto-reset.

STEP 7 OF 7 • SERVICE RATING

Help us improve municipal recycling! Select your rating from **1 to 5 Stars** using the keypad or on-screen touch.

اپنی رائے کا اظہار کریں: کی پیڈ سے اسے ۵ کا ہندسہ دیا کر سروس کا جائزہ لیں۔

★ STAR RATING KEYPAD SHORTCUTS

KEYPAD INPUT	RATING EQUIVALENT	CITIZEN SENTIMENT
Key 5	★★★★★	Outstanding Service (بہترین)
Key 4	★★★★	Good Experience (بہت اچھا)
Key 3	★★★	Satisfactory (تسلی بخش)
Key 2	★★	Needs Improvement (بہتری کی ضرورت)
Key 1	★	Unsatisfactory (غیر تسلی بخش)

Auto-Reset Sequence: After selecting a rating or after 10 seconds of inactivity, the kiosk displays a green checkmark thank you screen, flushes cache, and resets back to the Welcome Dashboard for the next citizen.

★
HOW WAS YOUR EXPERIENCE?
آپ کا ری سائیکلنگ کا تجربہ کیسا رہا؟

★★★★★

★★★★★ 5/5 — Excellent • بہترین

4. Exclusive PecoDrop 1.15x Eco Leaderboard

Wide-format landscape split screen featuring the expanded live citizen ranking engine.

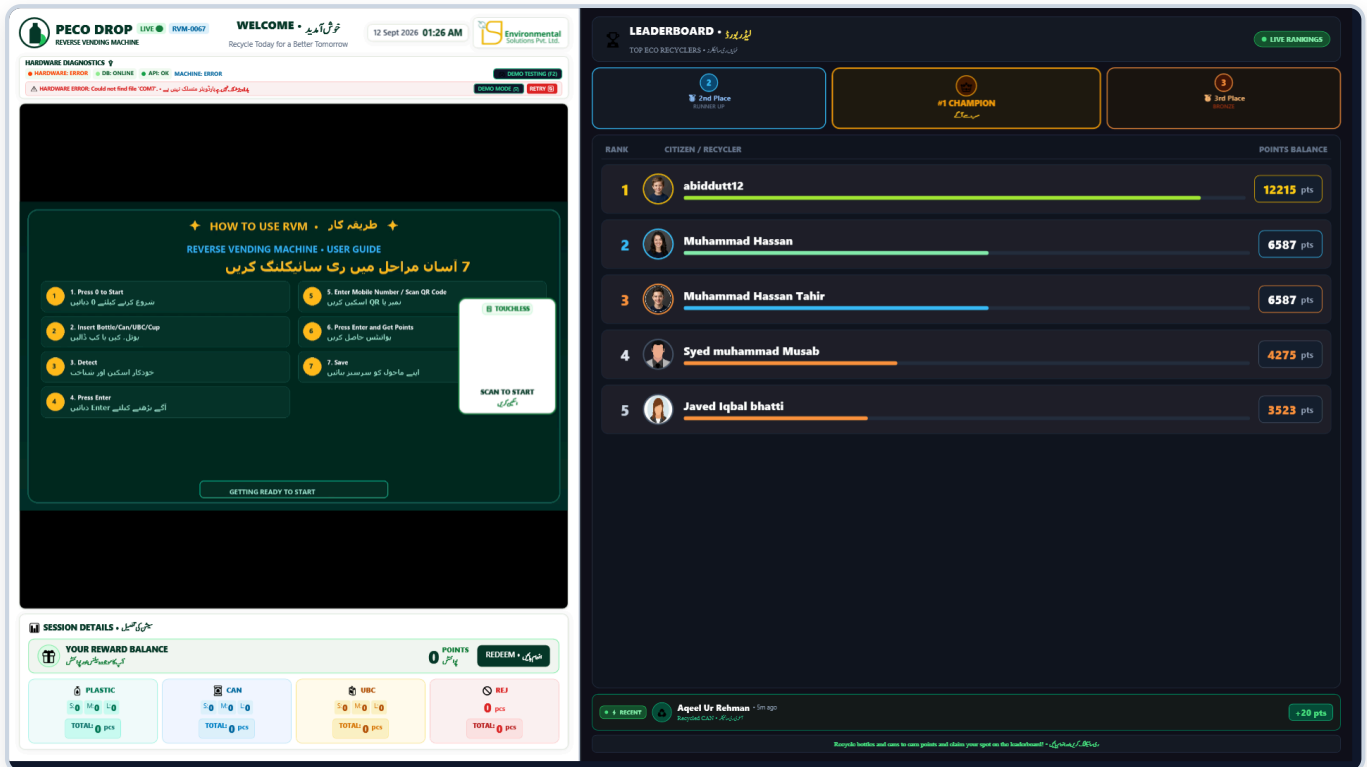


FIGURE 12.1 • PECODROP DUAL-ZONE LANDSCAPE KIOSK INTERFACE (LIVE CAPTURE)

Left Column (0.85x Width): Kiosk Centerpiece

- **High-Definition Instructional Video:** Continuous looping instructional animation demonstrating the perfect bottle, can, and tetra insertion.
- **Live Session Tally:** Compact live counter showing items inserted, current points accumulated, and session countdown timer.
- **Hardware Diagnostics:** Live telemetry showing COM7 Arduino link, SQL Server health, and cloud sync status.

Right Column (1.15x Width): Expanded Leaderboard

- **Top Citizen Recyclers:** Live ranking table displaying the city's premier recycling champions with real-time score updates.
- **Custom Avatars:** High-resolution citizen avatars (avatar1 to avatar5) bringing rich gamification to public spaces.
- **Points & Trophy Badges:** Highlights gold, silver, and bronze badges alongside total bottles recycled this month.
- **Community Impact:** Live stats showing total CO₂ offset and trees saved by the Katra Neem Wala community.

5. Hardware Keypad & Touchless Input Matrix

Comprehensive mapping of physical buttons, phone shortcuts, and touchless equivalents.

INPUT ELEMENT	SCREEN STATE / SCREEN MODE	ACTION TRIGGERED	TOUCHLESS ALTERNATIVE
Key '0'	Home / 1-Min Idle Screen	Initiates a new recycling session. Wakes machine from idle power save.	Scan on-screen QR Code.
Keys 1 – 9	Feedback / Rating Screen	Submits citizen satisfaction rating (1 = Unsatisfactory, 5 = Outstanding).	Tap rating star on screen.
Keys 0 – 9	Wallet Phone Number Screen	Enters citizen's 11-digit mobile phone number (03XXXXXXXX).	Scan dynamic wallet QR.
Key '#' (Enter)	Active Deposit Session	Finalizes the deposit phase and proceeds to points reward screen.	Tap 'Finish' button.
Key '#' (Enter)	Phone Entry Screen	Confirms mobile number and submits points to central wallet.	Auto-submits on 11 digits.
Key '*' (Clear)	Phone Entry Screen	Deletes the last typed digit (Backspace function).	Tap 'Clear' on display.
Key '*' (Cancel)	Active Deposit (0 items)	Cancels the session and returns kiosk to Welcome Dashboard.	Auto-cancels in 60 seconds.
Intake Light	All Operation Modes	Green = Ready to Insert Amber = Sizing Red = Reject/Locked.	On-screen visual cues.

 **PHYSICAL KEYPAD SPECIFICATIONS**

- **Material:** Vandal-proof stainless steel with engraved high-contrast numerals.
- **Feedback:** Tactile click with electronic audio beep upon every valid keypress.
- **Ingress Protection:** IP65 weather-sealed against rain, dust, and humidity.

 **TOUCHLESS QR SENSOR SPECIFICATIONS**

- **Dynamic Generation:** QR codes refresh every session with cryptographic timestamp.
- **Resolution:** High-contrast SVG rendering legible by all mobile cameras up to 2 meters.
- **Zero Contact:** Complete session can be performed without touching any surface.

6. Citizen FAQ & Troubleshooting

Answers to common recycling questions, error recovery, and citizen support helpline.

Q1: Why did the machine reject my plastic bottle? (میری بوتل مسترد کیوں ہوئی؟)

Answer: Common reasons include: (1) Residual liquid remaining inside, (2) Torn or missing barcode label, (3) Bottle heavily squashed or crumpled. Empty any liquid, reshape the bottle, and insert bottom-first.

Q2: Can I recycle beverage cans and tetra paks in PecoDrop?

Answer: Yes! Unlike basic reverse vending machines, PecoDrop features multi-material classification supporting Aluminum Cans (+15 points) and Tetra Pak juice cartons (+8 points) alongside PET bottles (+5 to +10 points).

Q3: What if I don't have a smartphone or the mobile app?

Answer: You do not need a smartphone! Start the session by pressing Key '0' on the keypad. When finished, enter your 11-digit mobile phone number on the keypad. Your reward points will be credited via SMS.

Q4: What happens if the internet goes offline during my deposit?

Answer: PecoDrop is equipped with offline failover. All points are securely stored in the local SQL Server database. As soon as the network reconnects, your points automatically sync to your cloud wallet.

Q5: How do I get featured on the 1.15x Eco Leaderboard?

Answer: Register your profile in the PecoDrop Mobile App, link your custom avatar, and deposit consistently. The city's top 5 recyclers are featured live on the kiosk's right-hand landscape screen!



Citizen Support & Assistance Helpline

Operating 24/7 for citizen inquiries, feedback, and machine maintenance requests.

0800-RECYCLE (73292)

support@isprvm.binishaqsoft.com